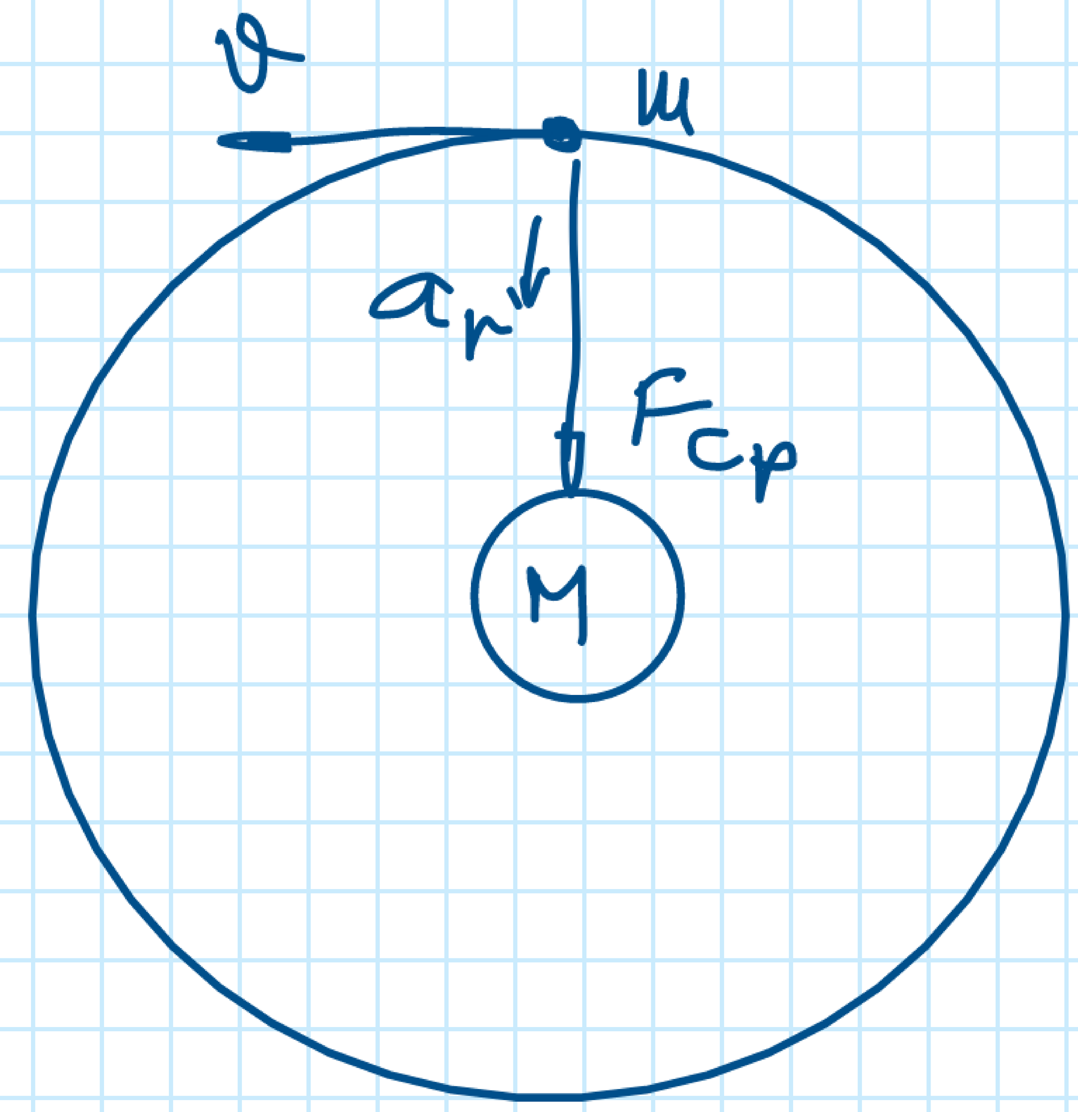
Ligjerata 7ç

Lëvizja e Tokës rreth Diellit



$$F = \gamma \frac{mM}{r^2}$$

$$F_{cp} = m \cdot a_r = m \frac{\vartheta^2}{r}$$

$$a_r = \frac{\vartheta^2}{r}$$

$$\vartheta = \omega \cdot r$$

$$F_{cp} = m \frac{\omega^2 r^{\cancel{2}}}{\cancel{r}} \; ; \; \omega = \frac{2\pi}{T}$$

$$\boxed{F_{cp} = m \frac{4\pi^2 r}{T^2}}$$

$$F = F_{cp}$$

$$\gamma \frac{\cancel{m}M}{r^2} = \cancel{m} \frac{4\pi^2 r}{T^2} \Rightarrow \frac{T^2}{r^3} = \frac{4\pi^2}{\gamma M} = \text{konst}$$

Ligjin e III të Keplerit

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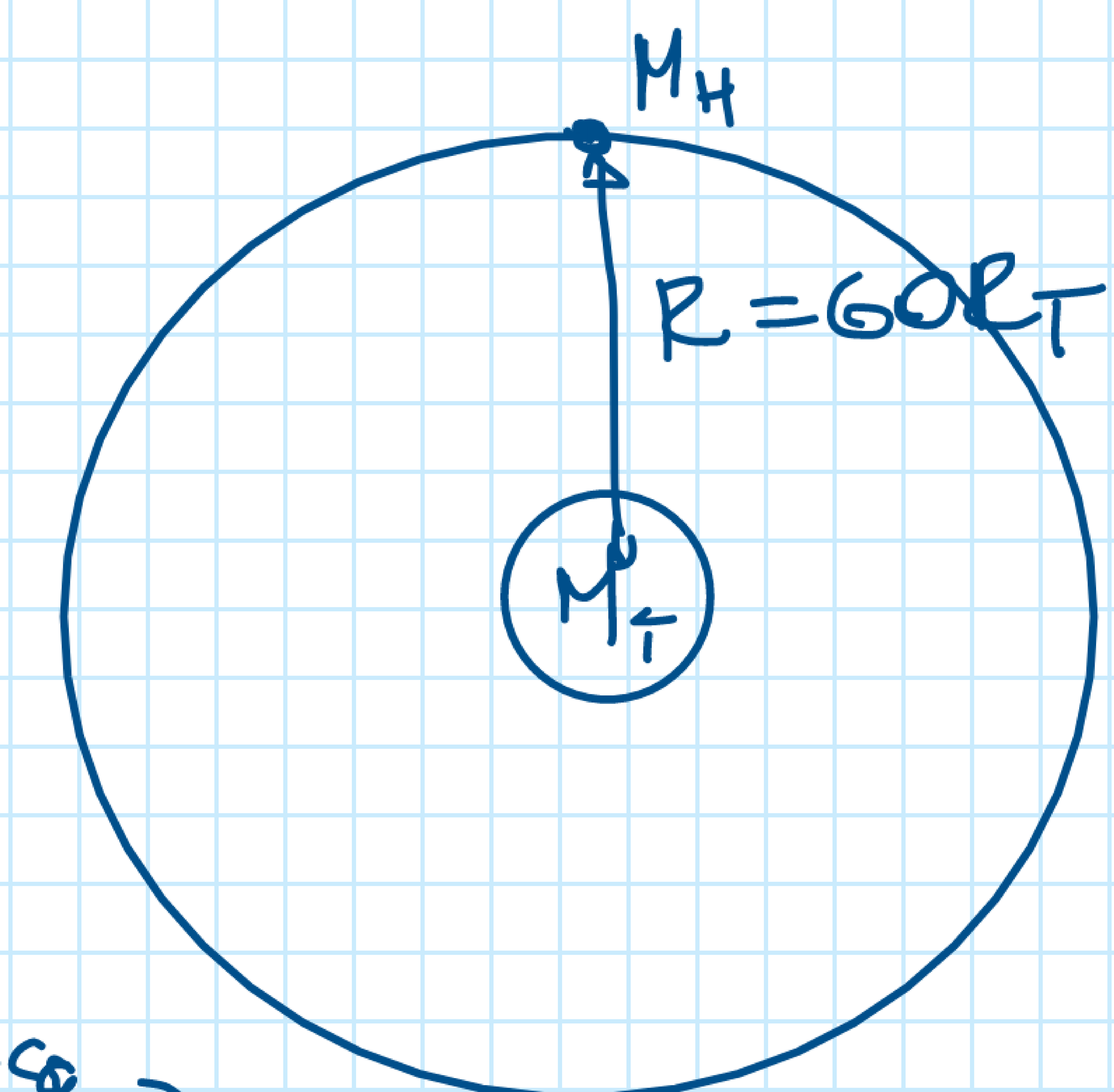
$$F = \gamma \frac{M_H M_T}{R^2}$$

$$F = M_H \cdot a \Rightarrow a = \frac{F}{M_H} = \frac{1}{\cancel{M_H}} \gamma \frac{\cancel{M_H} \cdot M_T}{R^2}$$

$$a = \gamma \frac{M_T}{R^2} \;\Big/\cdot \frac{R_T^2}{R_T^2}$$

$$a = \gamma \frac{M_T}{R_T^2} \frac{R_T^2}{R^2} \; ; \quad \boxed{g^0 = g = \gamma \frac{M_T}{R_T^2}}$$



$$a = g \frac{R_T^2}{R^2}$$

$$\left.\begin{aligned} R &= 384400 \text{ km} \\ R &\approx 60 R_T \end{aligned}\right\} \text{distanca Tokë-Hënë}$$

$$a = g \frac{\cancel{R_T^2}}{60^2 \cdot \cancel{R_T^2}} = \frac{9{,}81 \frac{m}{s^2}}{3600} = 2{,}7 \cdot 10^{-3} \frac{m}{s^2}$$